ASSIGNMENT 1

20INMCA501 - Data Science & Machine Learning

MCAINT2022-27-S9

Q1. Formulating a Research Goal, Project Charter, and Data Sources for Hospital Patient Data Analysis

1. Research Goal

The hospital seeks to analyze patient data to understand treatment patterns and improve the overall quality and efficiency of healthcare services delivered. The research goal can be formally stated as follows:

> *"To analyze historical and current patient treatment data in order to identify prevailing treatment patterns, evaluate treatment effectiveness, detect inefficiencies in patient care, and generate actionable insights that support improved clinical decision-making and healthcare service delivery."*

This goal aligns with the CRISP-DM "Business Understanding" phase, translating a broad organizational need into a specific, measurable, and data-driven objective.

2. Project Charter

A project charter formally defines the scope, objectives, and stakeholders of the analytics project. The key components are:

Component	Description
Project Title	Analysis of Patient Data for Improved Healthcare Service Delivery
Business Objective	Improve treatment outcomes, reduce inefficiencies, and support evidence-based clinical decisions.
Research Objective	Identify treatment patterns, correlations between patient attributes and outcomes, and areas for service improvement.
Scope	Includes patient demographics, diagnosis records, treatment history, and outcome data across departments; excludes billing and administrative systems.
Stakeholders	Hospital administration, doctors and clinical staff, data science team, IT/records department, patients (indirectly).
Deliverables	Cleaned dataset, exploratory analysis report, predictive/descriptive models, dashboards, and recommendations.
Timeline	Defined project phases: data collection, cleaning, exploration, modeling, and reporting, each with milestones.
Resources	Hospital database access, data science tools (Python/R), computing infrastructure, and domain expert support.
Constraints	Patient data privacy regulations (e.g., HIPAA/GDPR-equivalent norms), data quality limitations, and resource availability.
Risks	Incomplete or inconsistent records, ethical/legal risks in handling sensitive health data, resistance to adopting insights.

3. Appropriate Data Sources

To achieve the stated research goal, data can be retrieved from the following sources within and around the hospital ecosystem:

- **Electronic Health Records (EHR) / Electronic Medical Records (EMR):** Records containing patient demographics, medical history, diagnoses, and prescribed treatments.
- **Laboratory and Diagnostic Information Systems (LIS/RIS):** Detailed logs of laboratory tests, radiology results, and diagnostic imaging associated with each patient visit.
- **Hospital Management Information System (HMIS):** Data on ward occupancy, patient admission and discharge dates, and length of stay.

- **Pharmacy Records:** Records of medicines dispensed, dosage, and adherence, useful for studying treatment protocols.
- **Patient Feedback / Survey Data:** Surveys or feedback forms capturing patient satisfaction and perceived quality of care.
- **Staff and Resource Allocation Records:** Data on staff availability, specialization, and patient-to-doctor ratios that may influence treatment patterns.
- **External/Public Health Databases:** Publicly available datasets (e.g., WHO, government health portals) for benchmarking treatment patterns against regional/national standards.

These sources, once integrated, provide a comprehensive dataset enabling both descriptive analysis of current treatment patterns and predictive analysis for future service improvement.

Q2. Data Cleansing, Transformation, and Role of EDA for a Hospital Dataset

A hospital dataset containing missing values, duplicate records, inconsistent entries, and categorical variables requires systematic data cleansing and transformation before meaningful analysis can be performed.

1. Data Cleansing Techniques

(a) Handling Missing Values

- **Deletion:** Remove records/rows with missing values only when the proportion of missing data is very small and unlikely to bias results.
- **Imputation:** Replace missing numerical values (e.g., blood pressure, age) with mean, median, or mode; use domain-specific defaults or predictive imputation (e.g., KNN, regression) for more accuracy.
- **Forward/Backward Fill:** For time-based clinical readings, use forward-fill/backward-fill or interpolation techniques.

(b) Handling Duplicate Records

- **Deduplication:** Identify duplicates using unique patient identifiers (e.g., patient ID, admission number) combined with timestamp checks, and remove exact duplicate rows.
- **Record Linkage:** Merge records that refer to the same patient but were entered with slight variations (e.g., due to multiple registrations).

(c) Handling Inconsistent Entries

- **Standardization:** Standardize formats for dates, units of measurement (e.g., kg vs lbs), and text case for categorical fields (e.g., "Male/M/male" unified as "Male").
- **Validation Rules:** Identify outliers or invalid values (e.g., negative age, impossible vital signs) using range checks and statistical rules, then correct or flag them.
- **Fuzzy Matching:** Apply fuzzy matching to correct misspelled entries in diagnosis names, drug names, or department names.

(d) Handling Categorical Variables (Transformation)

- **One-Hot Encoding:** Convert nominal categories (e.g., blood group, gender, department) into binary columns suitable for modeling.
- **Label/Ordinal Encoding:** Convert ordinal categories (e.g., severity level: Low/Medium/High) into ordered numeric codes preserving rank.
- **Grouping Rare Categories:** For high-cardinality categorical fields (e.g., diagnosis codes), group rare categories together or use frequency/target encoding.

2. Role of Exploratory Data Analysis (EDA)

Once the dataset is cleaned and transformed, EDA is used to uncover useful patterns, relationships, and anomalies that guide further analysis and decision-making. EDA can be applied as follows:

- **Univariate Analysis:** Use histograms, box plots, and summary statistics (mean, median, standard deviation) to understand the distribution of variables such as age, length of stay, and treatment cost.
- **Bivariate/Multivariate Analysis:** Use scatter plots, correlation matrices, and cross-tabulations to study relationships between variables, such as the association between treatment type and recovery time, or age and disease severity.
- **Outlier Detection:** Box plots and z-score/IQR methods help identify outliers, such as unusually long hospital stays or abnormal vital signs, that may indicate data errors or critical medical cases.
- **Trend and Pattern Identification:** Line charts and time-series plots reveal seasonal patterns in admissions, disease outbreaks, or treatment demand over time.
- **Correlation Analysis:** Heatmaps and correlation coefficients help identify which factors (e.g., age, pre-existing conditions) are most strongly associated with treatment outcomes.

- **Segmentation:** Group-wise comparisons (e.g., by department, doctor, or treatment type) using bar charts and grouped statistics highlight differences in performance or resource utilization.

Through these techniques, EDA transforms the cleaned dataset into meaningful visual and statistical insights, enabling the hospital to identify inefficient treatment patterns, high-risk patient groups, and opportunities to improve overall healthcare service quality — directly supporting the research goal defined in Question 1.